

DEVIN THOMAS

PHD CANDIDATE • UNIVERSITY OF NEW HAMPSHIRE

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EDUCATION

UNIVERSITY OF NEW HAMPSHIRE

PhD in Computer Science, Advisor: Dr. Wheeler Ruml

2022-Dec 2026 (Expected)

Dissertation topic: Planning in a Dynamic World

UNIVERSITY OF BRITISH COLUMBIA

BSc with Combined Honours in Physics and Astronomy, Advisor: Dr. Gary Hinshaw

2012-2016

Thesis topic: Detecting Point Sources With CHIME Pathfinder

AWARDS

Best Paper Award, International Symposium on Combinatorial Search

2024

For our paper, Real-time Safe Interval Path Planning

JOURNAL PUBLICATIONS

Jordan G. Kueneman, Ernesto Bonadies, Devin Wild Thomas, David W. Roubik, and William T. Wcislo (2023). "Neotropical bee microbiomes point to a fragmented social core and strong species-level effects". *Microbiome*

Thomas H. Hampton, Devin Wild Thomas, Christopher van der Gast, George A. O'Toole, and Bruce A. Stanton (2021). "Mild cystic fibrosis lung disease is associated with bacterial community stability". *Microbiology Spectrum*

Laura C. Crane, Jason S. Goldstein, Devin Wild Thomas, Kayla S. Rexroth, and Alison W. Watts (2021). "Effects of life stage on eDNA detection of the invasive European green crab (*Carcinus maenas*) in estuarine systems". *Ecological Indicators*

CONFERENCE PUBLICATIONS

Devin Wild Thomas and Wheeler Ruml (2026a). "Complete Real-Time Planning in a Dynamic World". *Under Review*

Devin Wild Thomas, Ilya Atkin, and Thomas Butkiewicz (2026). "GapFiller 2.0: Web-based Planning Tools for Ocean Mapping in Support of Seabed 2030 Efforts". *OCEANS*

Devin Wild Thomas and Wheeler Ruml (2026b). "Learning, Commitment, and Completeness in Real-Time Heuristic Search". *Proceedings of the International Symposium on Combinatorial Search (SoCS)*

Issa Hanou, Eric Kemmeren, Devin Wild Thomas, and Mathijs de Weerd (2026). "Precomputing Multi-Agent Path Replanning Using Temporal Flexibility". *Proceedings of the International Symposium on Combinatorial Search (SoCS)*

Devin Wild Thomas and Wheeler Ruml (2025). "Real-time Cost-algebraic Heuristic Search". *Proceedings of the International Symposium on Combinatorial Search (SoCS)*

Devin Wild Thomas, Wheeler Ruml, and Solomon Eyal Shimony (2024). "Real-time safe interval path planning". *Proceedings of the International Symposium on Combinatorial Search (SoCS)*

Issa K Hanou, Devin Wild Thomas, Wheeler Ruml, and Mathijs de Weerd (2024). "Replanning in Advance for Instant Delay Recovery in Multi-Agent Applications: Rerouting Trains in a Railway Hub". *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS)*

Devin Wild Thomas, Tianyi Gu, Wheeler Ruml, and Eyal Solomon Shimony (2022). "Situating Grid Pathfinding Among Moving Obstacles (Extended Abstract)". *Proceedings of the International Symposium on Combinatorial Search (SoCS)*

WORKSHOP PUBLICATIONS

Devin Wild Thomas, Stephen Wissow, Michael Bauer, Paige McAfee, and Wheeler Ruml (2026). "Bounded-suboptimal Beam Search". *ICAPS 2026 Heuristics and Search for Domain-Independent Planning Workshop (HSDIP)*

Devin Wild Thomas, Solomon Eyal Shimony, Wheeler Ruml, Erez Karpas, Shahaf S Shperberg, and Andrew Coles (2023). "Any-Start-Time Planning for SIPP". *ICAPS 2023 Heuristics and Search for Domain-Independent Planning Workshop (HSDIP)*

UNREFEREED PUBLICATIONS

Devin Wild Thomas, Solomon Eyal Shimony, Wheeler Ruml, Erez Karpas, Shahaf S. Shperberg, and Andrew Coles (2026). *Optimal Planning in a Dynamic World*. arXiv: [2610.03312](https://arxiv.org/abs/2610.03312) [cs.AI]

Alison Watts, Laura Crane, Jason Garwood, Jason Goldstein, Megan Lamb, Christopher Peter, Yoshimi Rii, Shon Schooler, Devin Wild Thomas, William Thomas, et al. (2022). "Red fish, blue fish, native fish, new fish: eDNA as a tool to monitor fish in estuarine systems". *Preprint*

RESEARCH EXPERIENCE

UNIVERSITY OF NEW HAMPSHIRE

Durham, NH

2012-2014, 2018-Current

Research Assistant: 2025-Current

Research Group: Data Vis Lab at the Center for Coastal and Ocean Mapping

Project: Sonar Coverage Planning

Technologies: C++, Python, Coverage Planning, Cartographic Projections

Responsibilities:

Designed and implemented the planners for GapFiller 2.0, a web tool for sonar survey planning to support Seabed 2030.

Gap Filling Mode plans covering transits for polygonal unmapped regions

Transit Mode plans small changes to long transits to maximize new sonar coverage.

Deployed at: gapfiller.unh.edu

Research Assistant: 2022-2025

Research Group: UNH AI Group

Project: Planning in a Dynamic World

Technologies: C++, AI Planning, Heuristic Search, Real-time Search

Responsibilities:

Developed heuristic search-based approaches for planning in dynamic worlds.

Bioinformaticist: 2018-2022

Research Group: School of Marine Science and Ocean Engineering

Project: Aquatic Environmental DNA Detection

Technologies: Python, Bioinformatics, eDNA, Metabarcoding

Responsibilities:

Implemented analysis approaches for eDNA metabarcoding projects. Supported the National Estuarine Research Reserve System to set up early detection for invasive species. Tasks included data analysis, water sampling, and mentoring undergraduate researchers.

Summer Undergraduate Research Assistant: 2012-2014

Research Group: UNH Space Science Center

Project: ICME Analysis

Technologies: Python, Space Science

Responsibilities:

- Studied interplanetary coronal mass ejection data from the ACE satellite.

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

2013

Undergraduate Research Assistant

Project: Reagent Prep

Research Group: Moerman Lab

Technologies: Laboratory Methods

Responsibilities:

Prepared reagents and other minor tasks to aid the study of gene knockout in *C. elegans*.

TEACHING EXPERIENCE

UNIVERSITY OF NEW HAMPSHIRE

Durham, NH

2012-2014, 2017-2025

Teaching Assistant: 2023-2025

Department: Computer Science

Courses:

CS 730/830 – Introduction to Artificial Intelligence

CS 758/858 – Algorithms

CS 515 – Data Structures and Introduction to Algorithms

CS 419 – Computer Science for Engineers and Scientists

CS 415 – Introduction to Computer Science I

Bioinformaticist and Guest Lecturer: 2017-2022

Research Group: Hubbard Center for Genome Studies

Project: NH-INBRE Bioinformatics Curriculum

Technologies: Python, Bioinformatics

Responsibilities:

Helped develop and implement the NH-INBRE bioinformatics curriculum, supporting the introduction of modern 'big-data' bioinformatics to undergraduate students.

Guest Lecturer at Dartmouth, University of New Hampshire, Bates, Colby-Sawyer, Keene State College, New England College, Plymouth State University, and Saint Anselm College.

Workshop Instructor for:

Train the Trainers Workshop 2018-2022

Mount Desert Island Biological Lab

Benthic Invertebrate Taxonomy, Metagenomics, and Bioinformatics 2017-2018

Harte Institute, TAMU-CC

Instructor: 2017

Department: Genetics

GEN 712/812 – Programming for Bioinformatics a hybrid undergraduate/graduate course on using Python programming for genomics research. Co-taught with a Ph.D. student, splitting work 50/50; duties included developing course material, lecturing, and grading.

Project SMART Mentor 2012-2014

Department: UNH Space Science Center

- Mentored high school students with their Project SMART research projects.

PHILLIPS EXETER ACADEMY

Exeter, New Hampshire

2023

Long-Term Substitute Instructor

Department: Computer Science

Took over instruction of three sections of two introductory CS courses.

- Successfully resumed instruction after disruption of short-term substitute and uncertain return of original instructor.
- Restarted Harkness discussions on current topics in computer science, such as E-Waste, generative AI, and the ethics of self-driving cars.

PROFESSIONAL ACTIVITIES

Organizing

- ICAPS Workshop on Heuristics and Search for Domain-Independent Planning 2025, 2026

Journal Reviewing

- Artificial Intelligence

Conference Reviewing

- International Conference on Planning and Scheduling
- International Symposium on Combinatorial Search
- Workshop on Graph-Theoretic Concepts in Computer Science

Workshop Reviewing

- Workshop on Heuristics and Search for Domain-independent Planning